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Digitální zpracování obrazu

Lecture 3

Digitální zpracování obrazu

- Digitalizace (vzorkování, kvantování)
- Předzpracování (změny kontrastu a jasu, potlačení šumu, zaostření, ...)
- Analýza obrazu (detekce a rozpoznávání objektů)
- Kódování (komprese)

Předzpracování obrazu

- **Image enhancement**
- **Image restoration**

Image enhancement

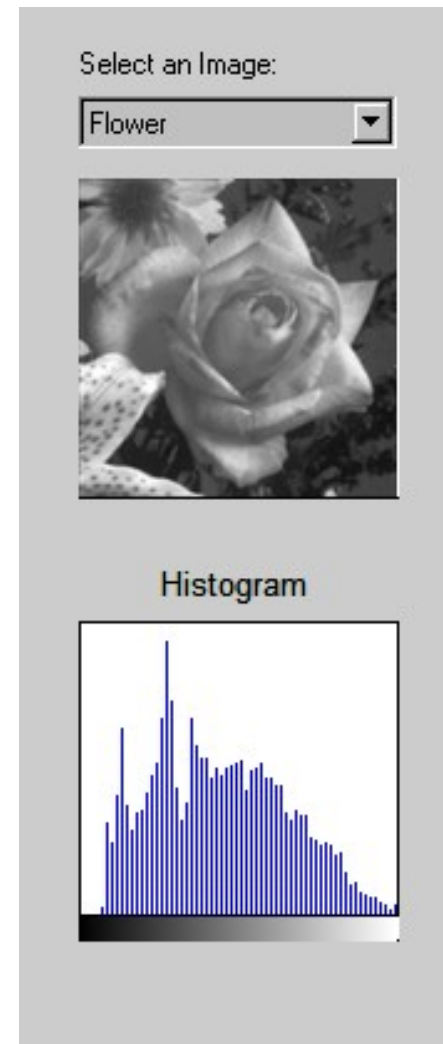
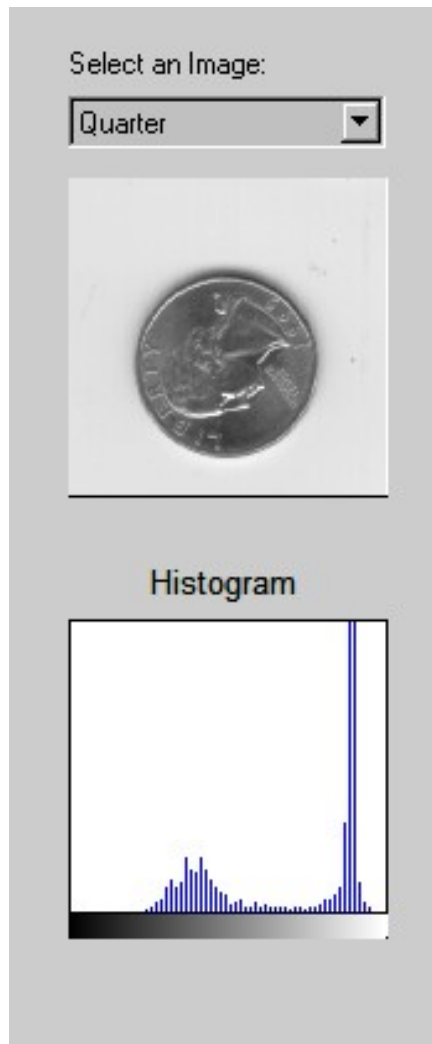
- **Změny kontrastu a jasu**
- **Potlačení šumu**
- **Detekce a zvýraznění hran**



Změny kontrastu a jasu

- **Histogram**
- **Lineární změny kontrastu a jasu**
- **Nelineární změny kontrastu a jasu, ekvalizace**
- **Lokální operace**

Histogram

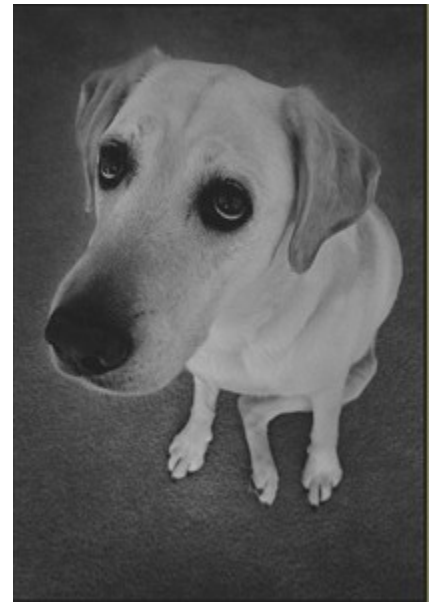




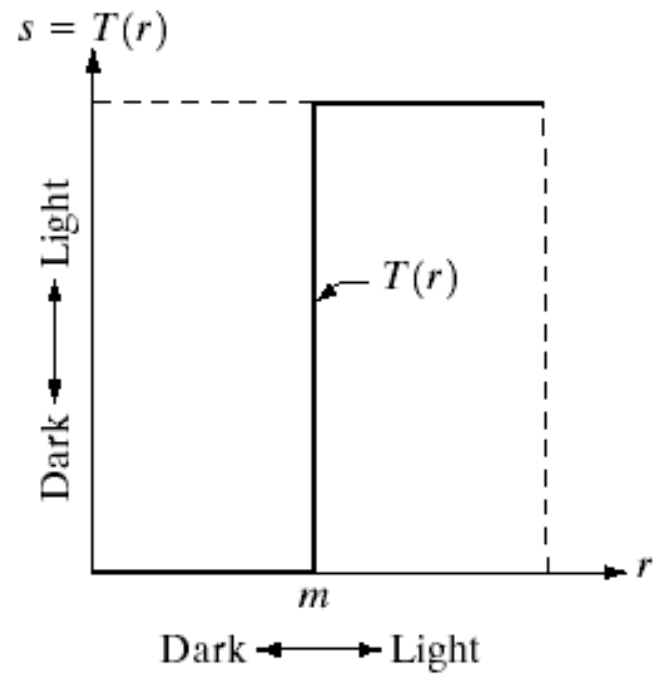
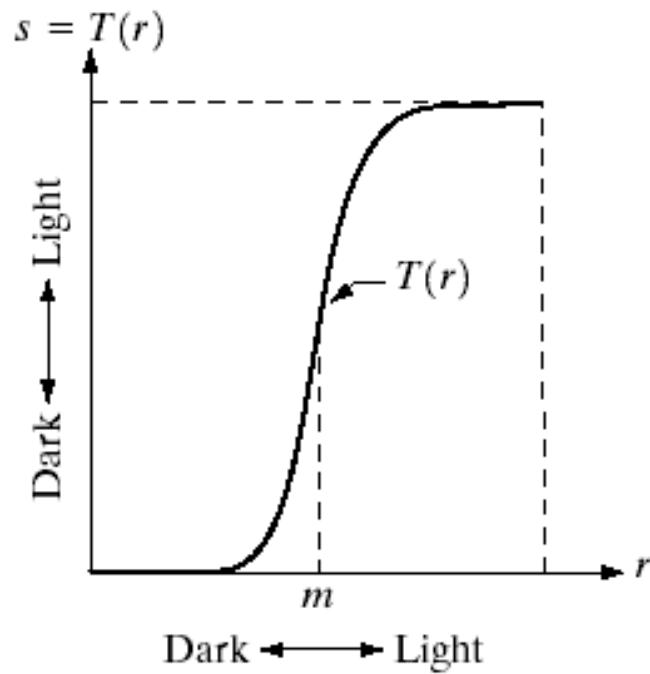
jas



kontrast

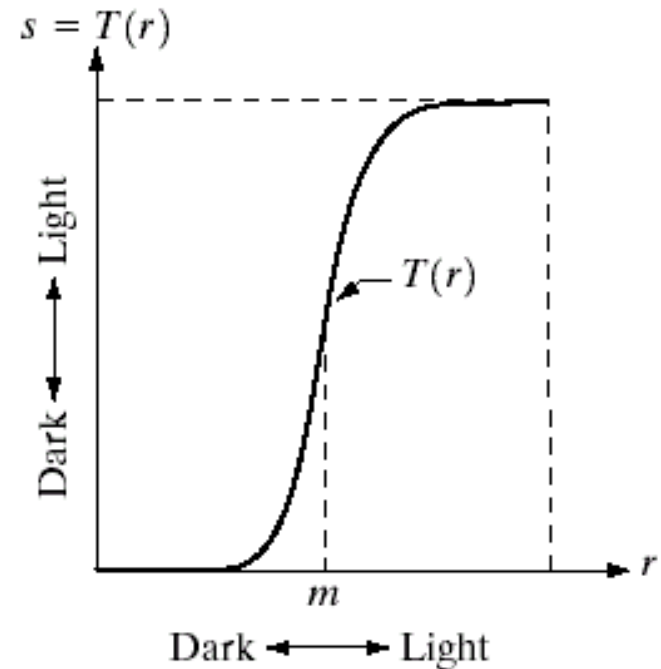


Změny kontrastu a jasu



Categories of illumination changes

- Point-wise transforms

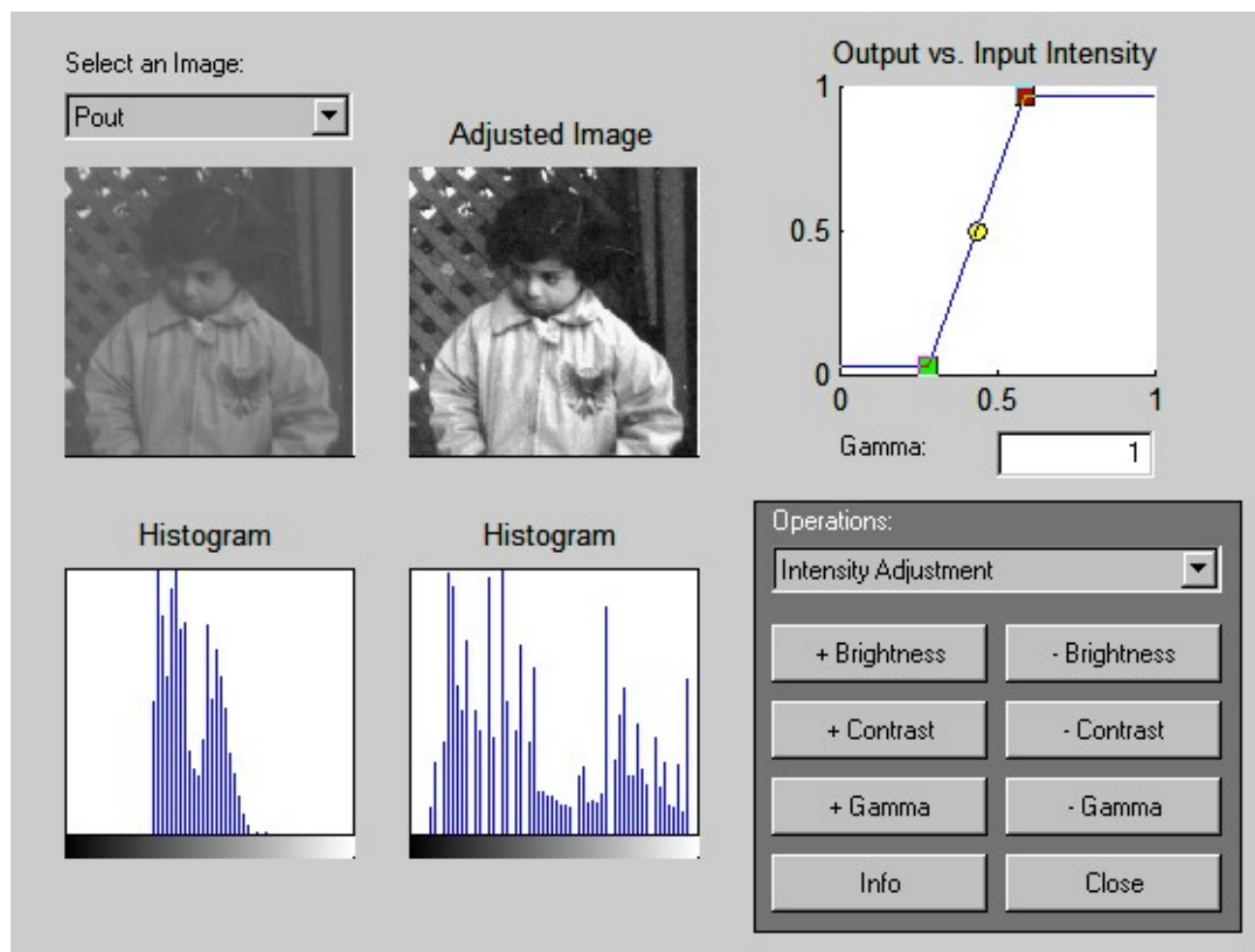


- Global x local
- Linear x nonlinear

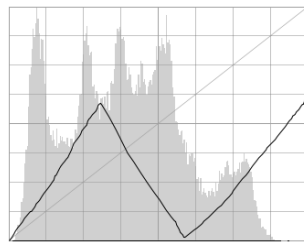
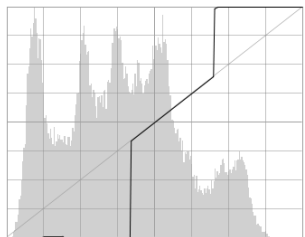
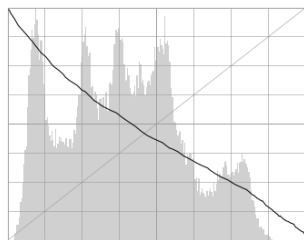
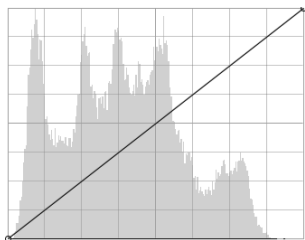
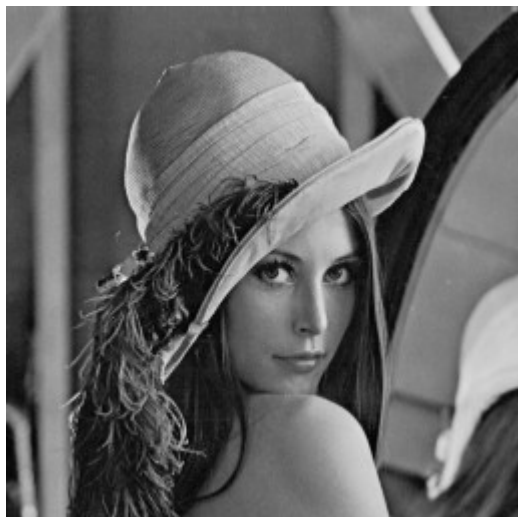
Illumination changes



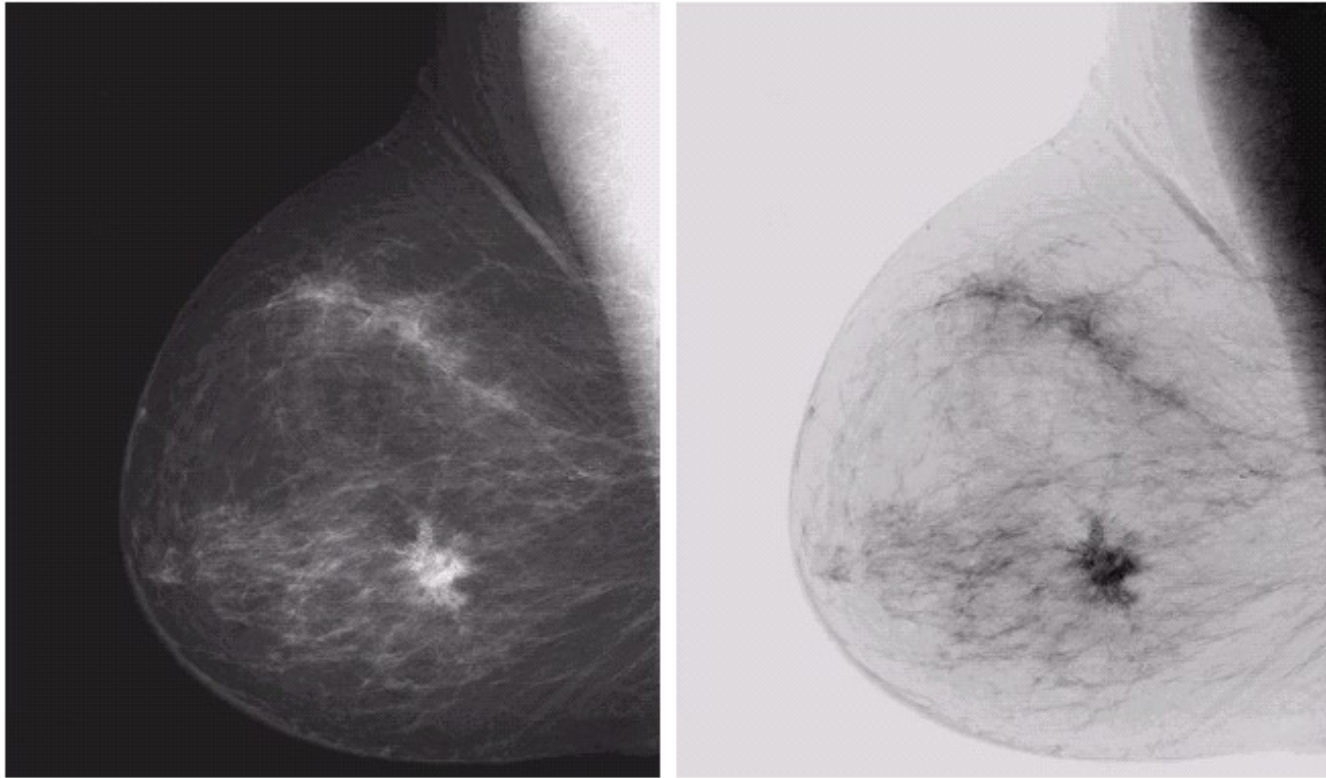
Změny kontrastu a jasu – lineární





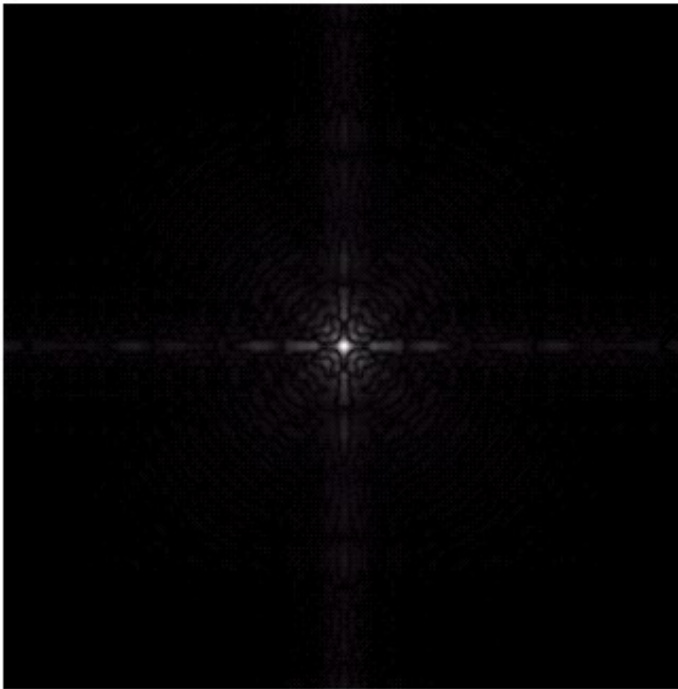


Negativ

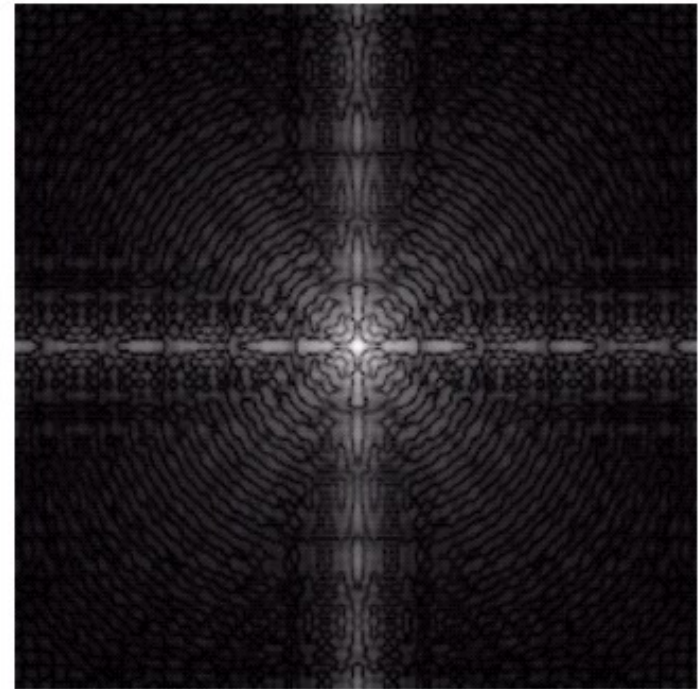


Mamogram

Nelineární transformace šedi



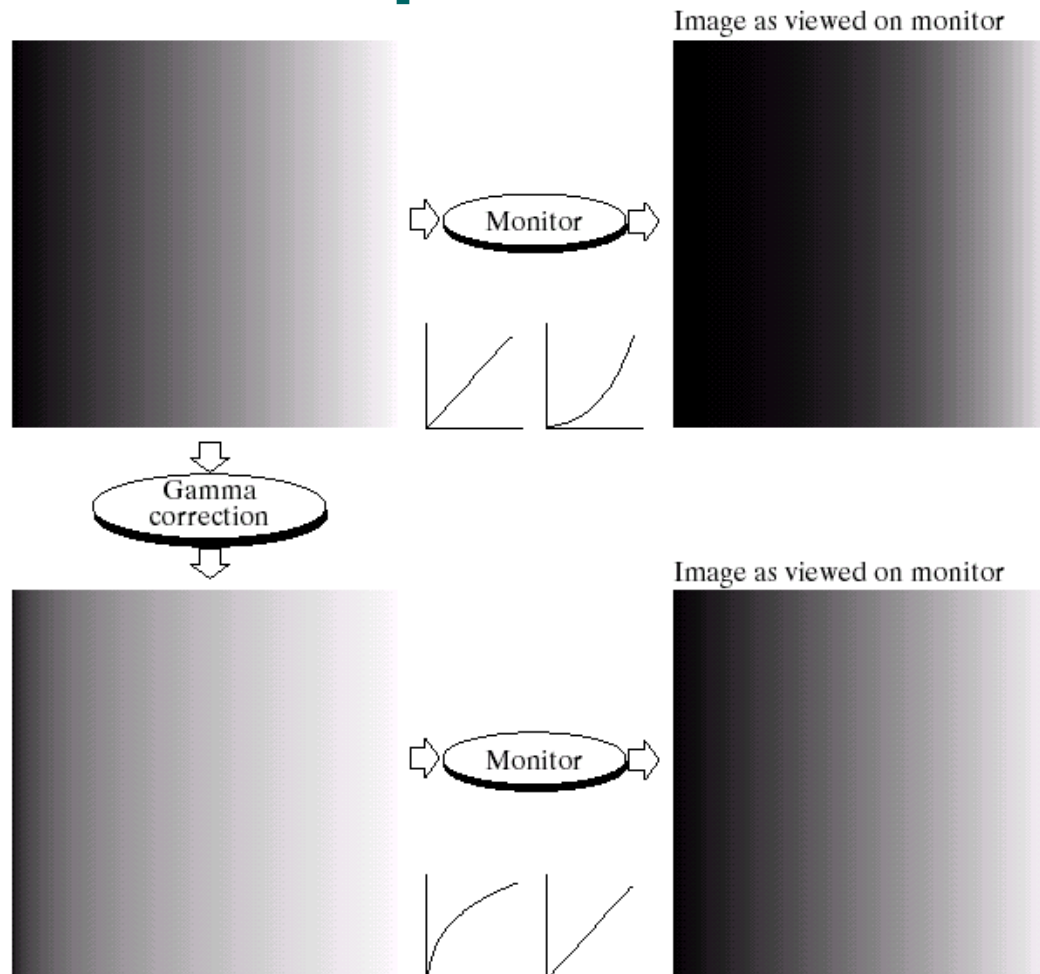
ampl



$\log(\text{ampl} + 1)$

Gama korekce

$$\text{Output} = c (\text{input})^{\text{gama}}$$



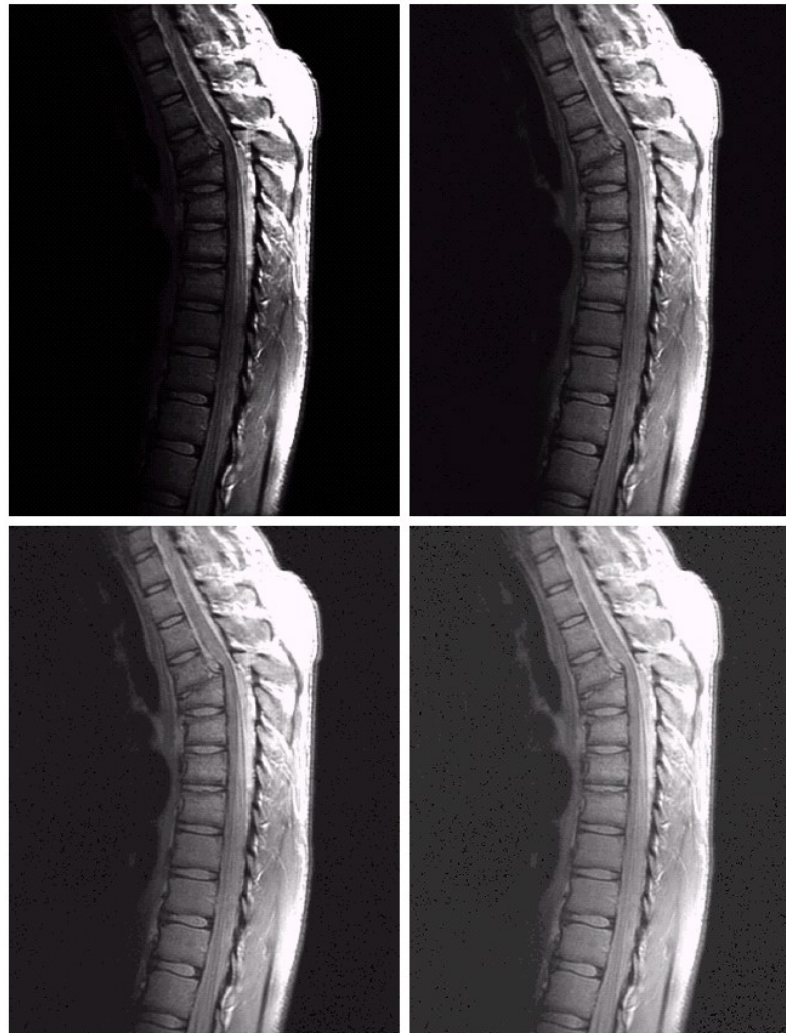
Gama korekce

Output = (input)^{gama} gama = 3, 4, 5

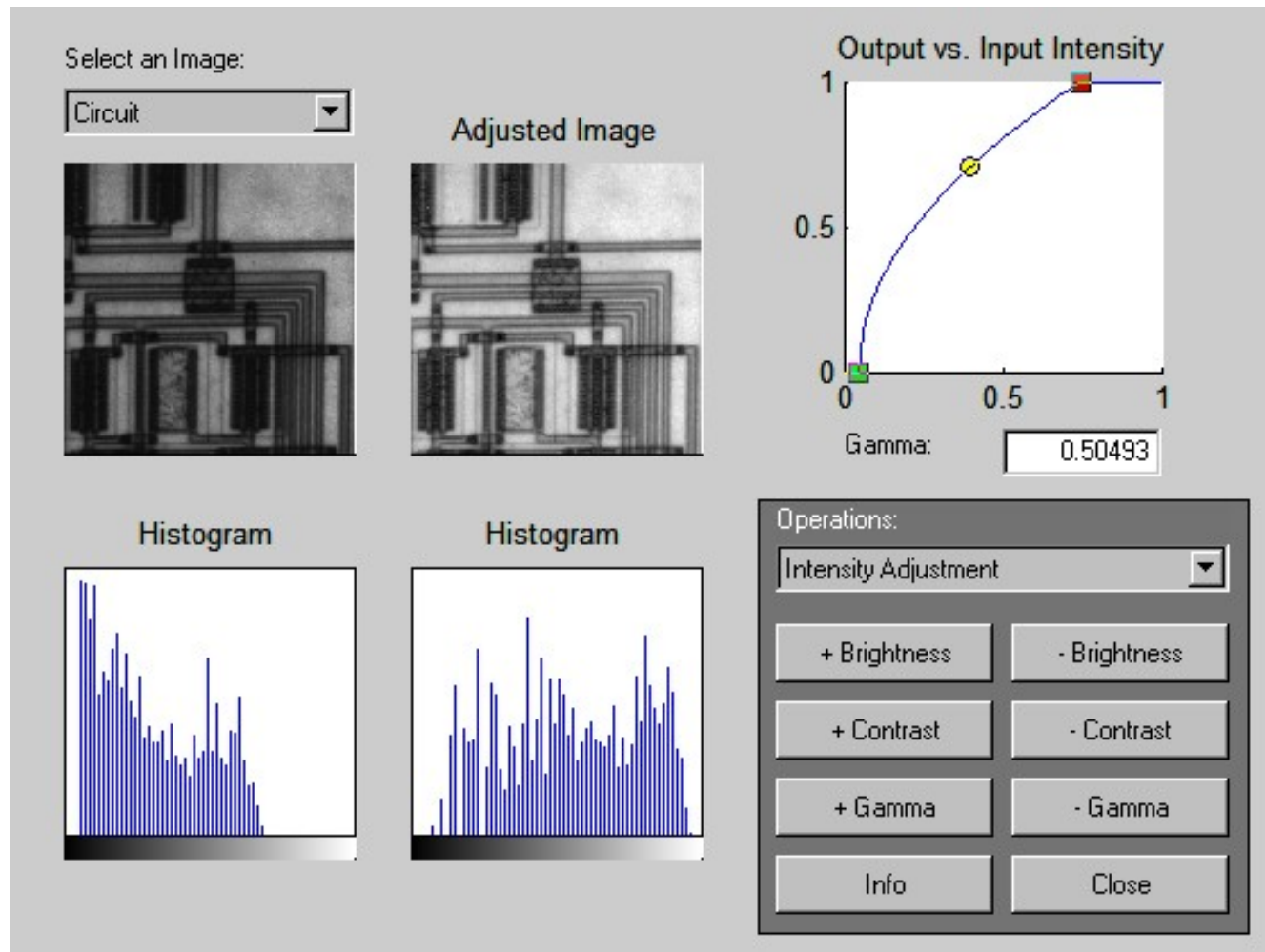


Gama korekce

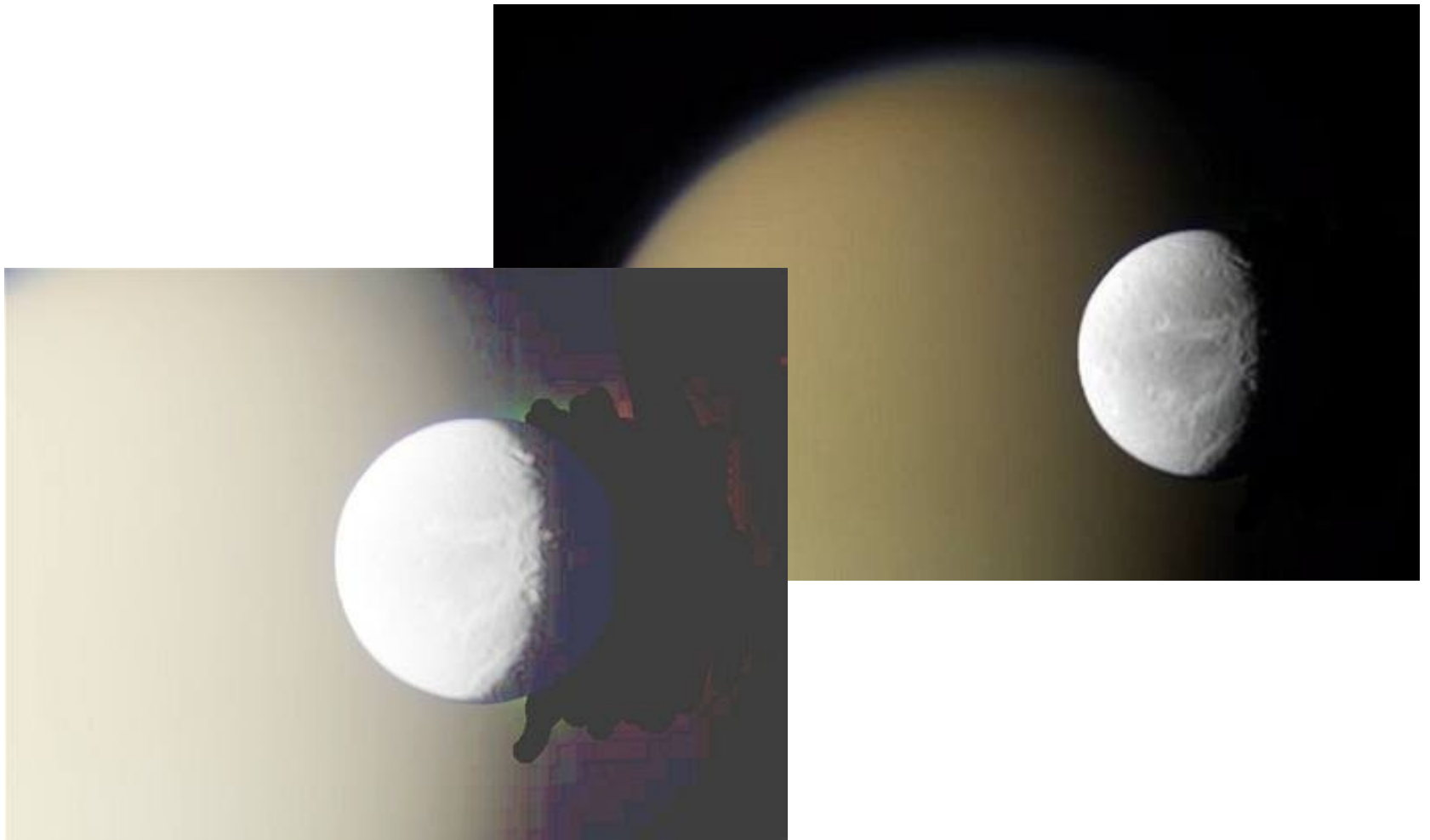
Output = (input)^{gama} gama = 0.6, 0.4, 0.3



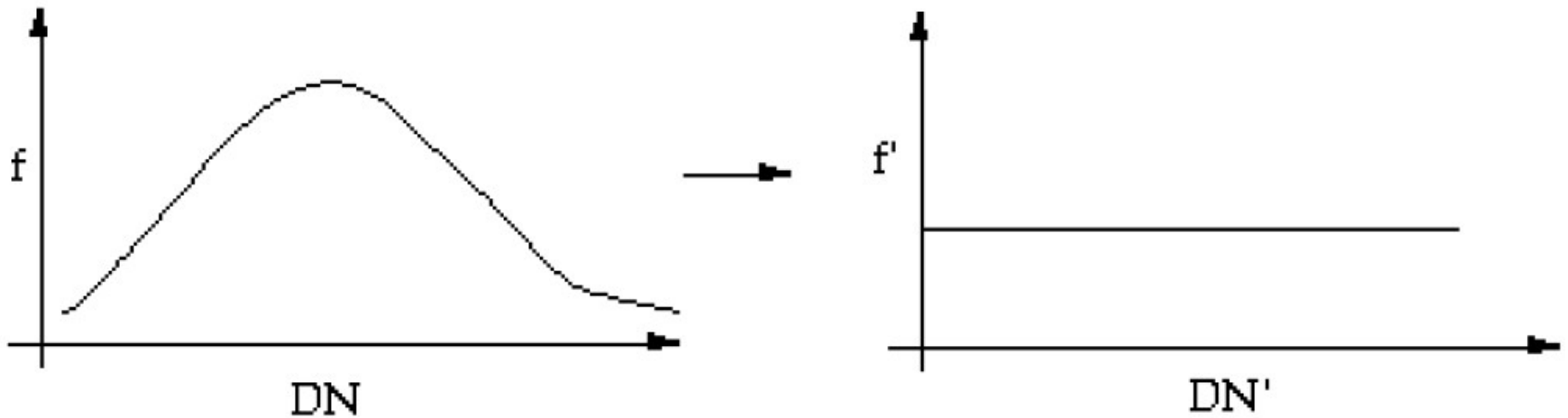
Gama korekcije



Gama korekce



Ekvalizace histogramu

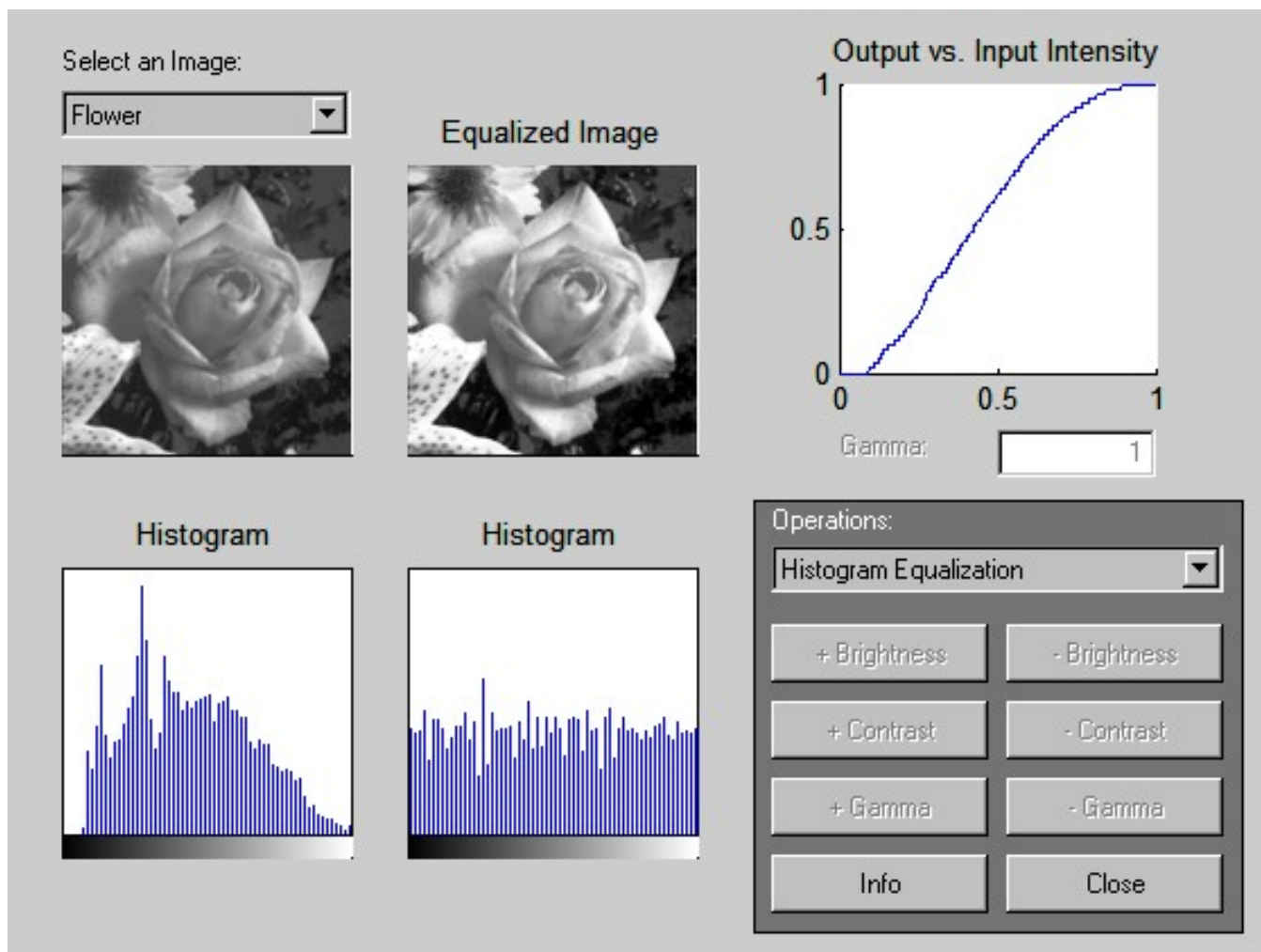


řešení -> kumulativní histogram

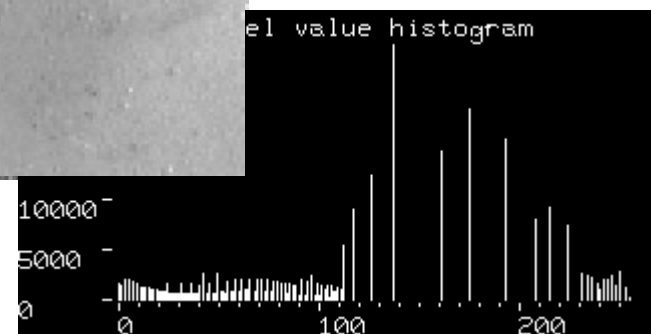
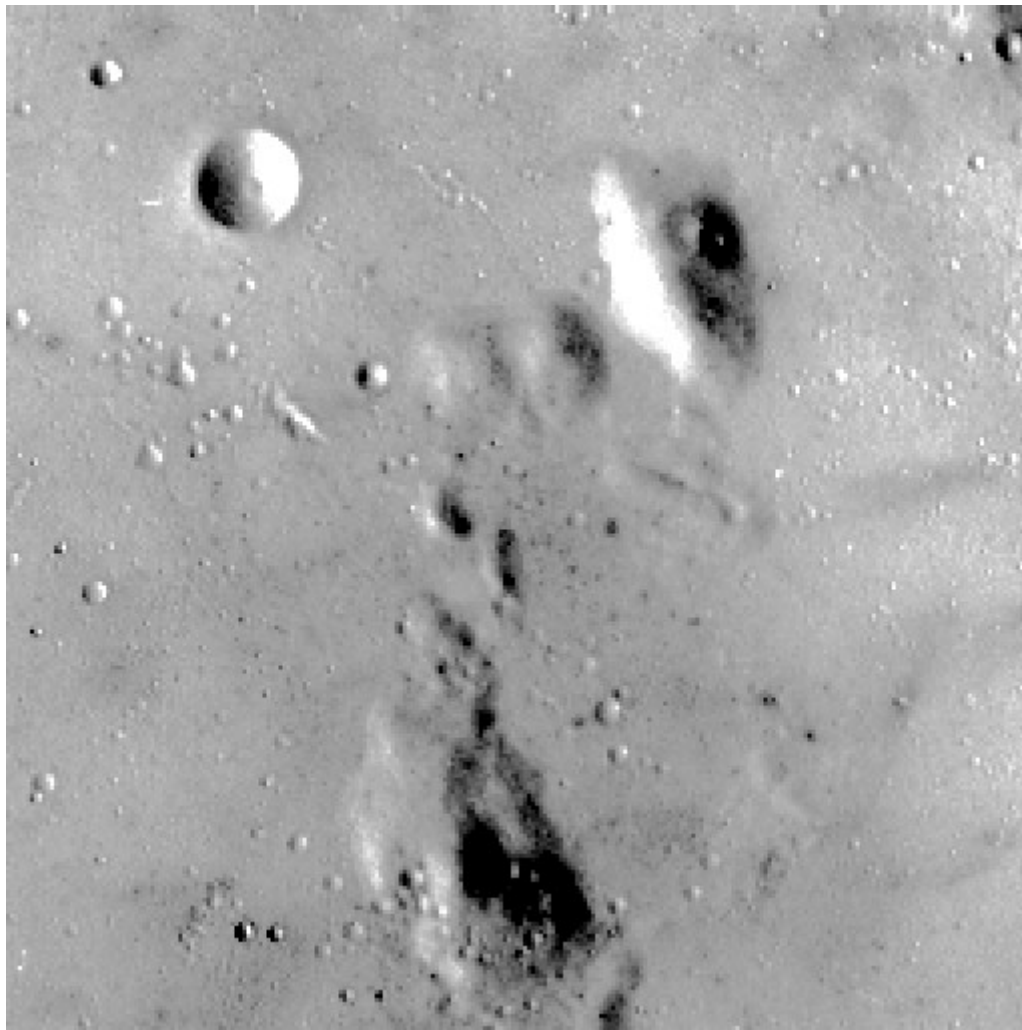
Ekvalizace histogramu

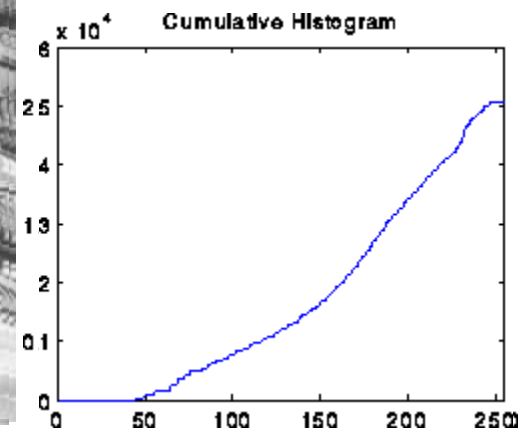


Ekvalizace histogramu



Ekvalizace histogramu





**Díky, pro dnešek končíme s
úpravami kontrastu!**

Nějaké otázky ?